Kanghwi Lee

Ph.D. Candidate | Audio Machine Learning and Computational Neuroscience

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Research Profile

Applied ML researcher and ETH Zurich Ph.D. candidate developing scalable models and data workflows for real-world audio and biological signals. Research spans representation learning, foundation model adaptation, large-scale audio data curation, and generative modeling.

Selected Research

Large-Scale Audio Representation Workflow and Human-in-the-Loop Clustering

First-author paper, Applied Sciences, 2026 | Paper | Code

- Developed a semi-automated workflow using Whisper encoder embeddings, UMAP-based clustering, and interactive expert review to organize real-world vocalization data.
- Achieved ARI = 0.73 and NMI = 0.81 on a 1,454-vocalization benchmark while reducing annotation time by approximately 50% (90 to 45 minutes per bird).
- Applied the workflow to approximately 80,000 hours of recordings for downstream doctoral research.

WhisperSeg: Long-Context Voice Activity Detection for Human and Animal Recordings

Co-authored paper, ICASSP, 2024 | Paper

- WhisperSeg adapts a pretrained Whisper Transformer for long-context, token-based voice activity detection, generating onset, offset, and vocalization-type tokens directly from audio spectrograms across human and animal recordings.
- Achieved segment-wise F1 = 0.9966 on adult zebra finch recordings and outperformed frame-wise baselines in segment-wise evaluation across all tested datasets, including noisy marmoset and human recordings.
- Developed dataset-processing and baseline-evaluation components for the study, then applied WhisperSeg segmentation outputs in subsequent large-scale vocalization analysis workflows.

Counterfactual Audio Generation and Trajectory Variance

Submitted for review to Interspeech 2026 | Code

- Designed and implemented a PyTorch-based age-conditioned counterfactual model using VAE latent representations and mini-batch optimal-transport matching, with single-forward-pass latent displacement prediction.
- Evaluated on approximately 680K vocalizations across three zebra finches, outperforming Gaussian OT, per-age k-NN, and per-age OT baselines; Cohen's $d = 0.29$–0.57 and AUC = 0.58–0.67 after controlling for duration.

Computational Modeling of Vocal Template Formation

COSYNE, 2024 | Conference Abstract

- Developed a Bayesian causal-inference model of composite song-template formation from sequential tutoring, fitted to behavioral data from 19 juvenile zebra finches (26 syllables).

Publications and Conference Abstracts

Journal and Conference Papers

- **Kanghwi Lee**, Maris Basha, Anja T. Zai, and Richard H. R. Hahnloser. "Benchmarking Automated and Semi-Automated Vocal Clustering Methods." *Applied Sciences*, 16, 810, 2026. DOI
- Nianlong Gu, **Kanghwi Lee**, Maris Basha, Sumit Kumar Ram, Guanghao You, and Richard H. R. Hahnloser. "Positive Transfer of the Whisper Speech Transformer to Human and Animal Voice Activity Detection." *ICASSP*, 2024. DOI

Manuscript Submitted for Review

- **Kanghwi Lee**. "Trajectory Variance: An Unsupervised Measure of Developmental Vocal Plasticity." Submitted for review to *Interspeech 2026*.

Conference Abstract

- **Kanghwi Lee**, Richard H. R. Hahnloser, Hazem Toutounji, and Dina Lipkind. "Serial Tutoring Reveals a Composite Song Template." *COSYNE*, 2024.

Education

ETH Zurich – Ph.D. Candidate, Institute of Neuroinformatics | *Apr 2022 – Present; expected early 2027*

University of Zurich – M.Sc., Neural Systems and Computation | *Feb 2020 – Feb 2022; GPA 5.5/6.0*

Korea University – B.S., Computer Science; B.Econ., Statistics | *Mar 2014 – Feb 2020; GPA 4.02/4.50*

Technical Skills

ML and Generative Modeling: PyTorch, Transformers, Whisper, representation learning, VAEs, flow matching, diffusion models, optimal transport

Audio and Data Workflows: audio segmentation/VAD, learned audio representations, spectrogram analysis, clustering, interactive annotation, large-scale audio processing

Programming and Computing: Python, NumPy/SciPy, scikit-learn, librosa, MATLAB, Git, Linux, HPC, parallel computing, GPU-based batch processing